

PWOKA TECHNOLOGIES LTD.

ENTERPRISE NETWORK DESIGN PROPOSAL

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Unit Name:	CCNA3
Project: Design	Multi-Site Enterprise Network Infrastructure
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1. INTRODUCTION

This Network Design Proposal outlines the strategic planning, architectural layout, and logical structure of a modern enterprise network infrastructure for Pwoka Technologies Ltd. The purpose of this proposal is to establish a secure, scalable, and highly available communication platform that bridges a multi-departmental Head Office campus with a geographically isolated Branch Office.

Prior to physical deployment and device configuration within Cisco Packet Tracer, this document serves as the formal engineering blueprint. It translates high-level business goals into a structured, deterministic technical layout that aligns with Cisco enterprise network design methodologies.

2. PROJECT OBJECTIVES & BUSINESS REQUIREMENTS

2.1 PROJECT SCOPE & OBJECTIVES

The overarching objective of this design project is to engineer a highly available, secure, and easily expandable enterprise network infrastructure. This system will integrate the logical workflows of Pwoka Technologies Ltd. across two main physical sites. The design scope explicitly encompasses:

- Designing a physical topology consisting of a multi-switch Head Office and a distinct Branch Office.
- Engineering a logical VLAN segmentation framework to isolate department broadcast domains.
- Establishing redundant Layer 3 gateway paths and high-capacity switch aggregation links.
- Creating a deterministic IPv4 addressing matrix spanning internal local area networks and wide area network (WAN) point-to-point links.
- Designing a security baseline to protect device access lines and control local network behaviour.

2.2 BUSINESS REQUIREMENTS

To properly serve the corporate needs of Pwoka Technologies Ltd., the proposed infrastructure must satisfy the following organizational criteria:

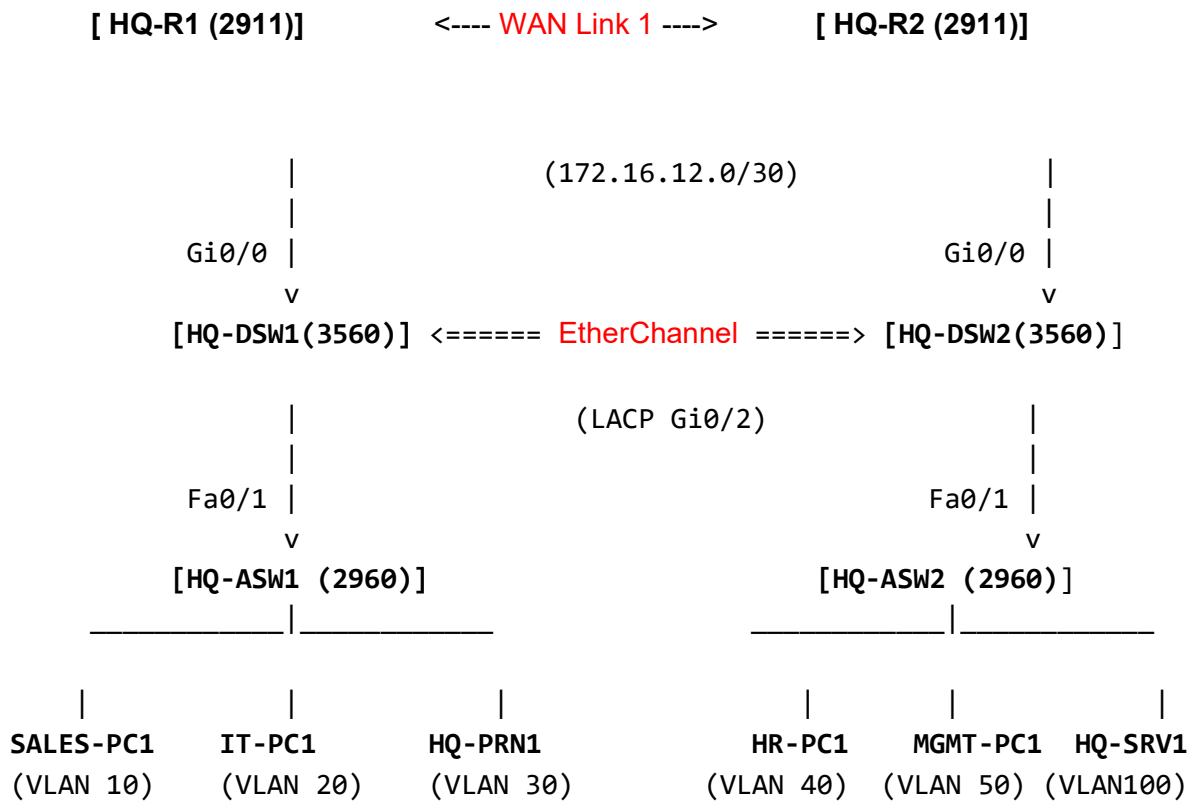
- **Multi-Site Connectivity:** The network must provide reliable communication pathways between the localized Head Office workforce and the remote Branch Office team.
- **Functional Department Isolation:** Individual business units must be cordoned off into separate security and broadcast zones to minimize cross-departmental exposure.
- **High Availability & Fault Tolerance:** The design must survive single cable drops or device-level hardware failures without causing corporate network downtime.
- **Asset Security & Remote Management Access:** All administration pathways must be fully encrypted to safeguard the integrity of internal network data.
- **Scalability for Future Expansion:** The logical addressing structure must support seamless expansion, allowing the company to add new users, hardware, or branch offices later without requiring a redesign.

The physical design for Pwoka Technologies Ltd. is structured as a dual-site architecture

The physical design for Pwoka Technologies Ltd. is structured as a dual-site architecture. The Head Office acts as the central data and routing hub, while the Branch Office operates as a lean, remote operational spoke. The two sites are securely connected across geographical distances using high-speed point-to-point serial WAN links.

HEAD OFFICE CAMPUS

=====



=====

BRANCH OFFICE

=====

[HQ-R2 (2911)]

|

| **WAN Link 2** (172.16.23.0/30)

v

[BR-R1 (2911)]

| Gi0/0

v

[BR-ASW1 (2960)]

|

|
BR-PC1

|
BR-PC2

|
BR-PC3

|
BR-PRN1

----- VLAN 40 ----- (VLAN 30)

3.2 VISUAL ASSET INTERGRATION

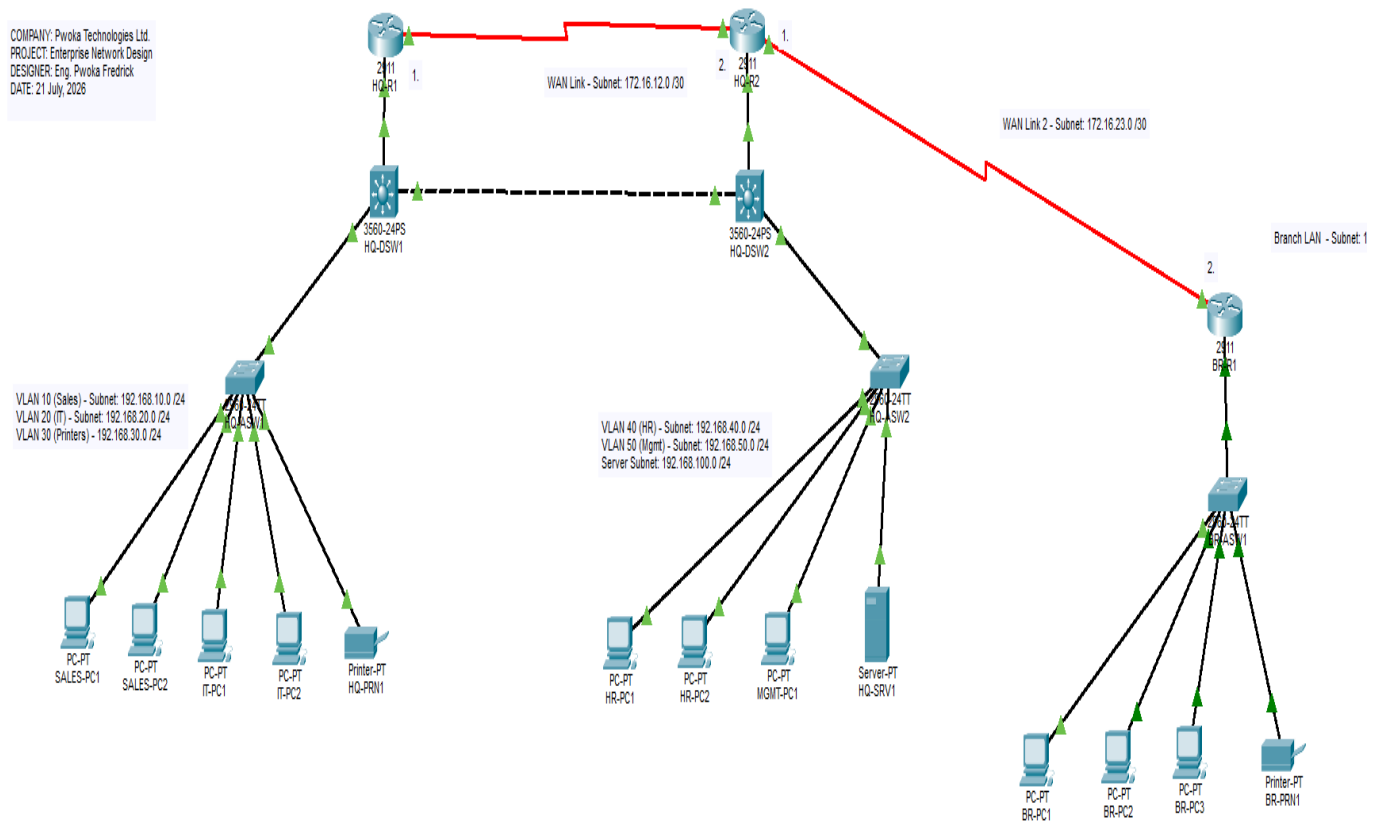


Figure 1: Proposed Enterprise Network Topology Blueprint for Pwoka Technologies Ltd.

4. HARDWARE SELECTION MATRIX

Device Type	Selected Model	Role in Topology	Engineering Justification
Enterprise Router	Cisco 2911 ISR	Core Routing & WAN Aggregation	Features three onboard 10/100/1000 Mbps interfaces and high-density modular slots. Supports HWIC serial cards, EIGRP, and HSRP engine processing natively.
Multilayer Switch	Cisco 3560-24PS	Core Distribution Layer	Operates at both Layer 2 and Layer 3. Features wire-speed hardware routing engines. Supports Switch Virtual Interfaces (SVIs) and LACP EtherChannel bundles.
Access Switch	Cisco 2960-24TT	Edge Layer Connectivity	Provides 24 Fast Ethernet access ports for end-user nodes with dedicated Gigabit uplinks. Fully supports 802.1Q tagging, STP, and port security.

5. HIERACHICAL NETWORK DESIGN MODEL

5.1 THE CISCO THREE-TIER PARADIGM

The proposed infrastructure avoids a traditional flat network deployment in Favor of the **Cisco Hierarchical Network Design Model**. Cisco recommends this structured approach because it breaks down the network into three distinct functional layers: Core, Distribution, and Access. This modular system ensures that a failure at one point does not bring down the entire network, simplifies administrative isolation, and allows for seamless future upgrades.

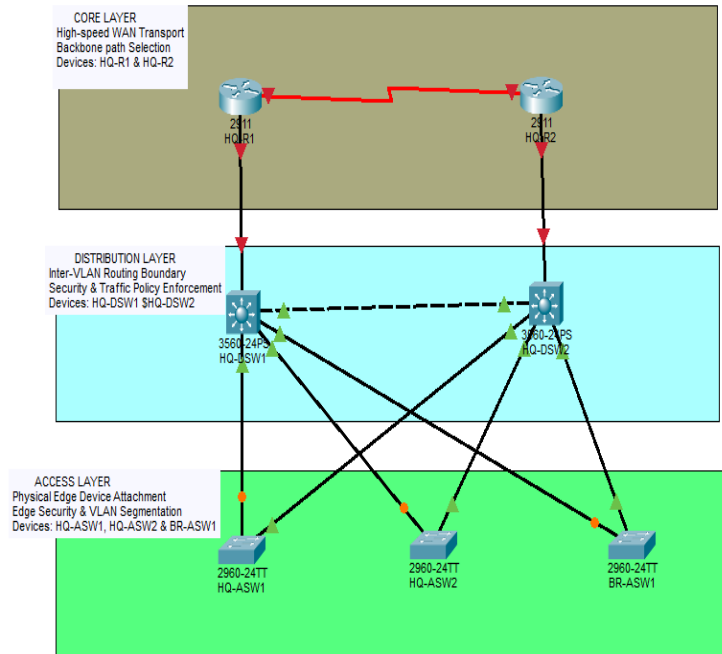


Figure 2: Hierarchical Network Model Architecture Layout

5.2 LAYER IMPLEMENTATION ANALYSIS

The proposed blueprint implements this tiered structure across the following specific organizational boundaries:

- The Core Layer:** Handled by the **Cisco 2911 Routers (HQ-R1, HQ-R2, and BR-R1)**. This layer focuses on fast, reliable transport of data across wide area network (WAN) serial links, moving packets between geographical sites without processing internal VLAN tags.

- **The Distribution Layer:** Managed by the **Cisco 3560 Multilayer Switches (HQ-DSW1 and HQ-DSW2)**. This layer bridges the routing boundary and the access ports. It routes data between local subnets, applies traffic rules, and provides backup paths for user gateways.
- **The Access Layer:** Managed by the **Cisco 2960 Switches (HQ-ASW1, HQ-ASW2, and BR-ASW1)**. This layer provides physical plugin ports for user computers, network printers, and servers. Its primary job is to assign incoming data to its correct VLAN right at the network edge.

6. VLAN SEGMENTATION ARCHITECTURE

6.1 ARCHITECTURAL DESIGN JUSTIFICATION

The logical topology for Pwoka Technologies Ltd. rejects a traditional flat network infrastructure in Favor of an optimized Virtual Local Area Network (VLAN) segmentation design. In a flat network, all hosts reside within a single broadcast domain, which exposes the enterprise to severe performance degradation from broadcast storms and presents massive security vulnerabilities.

The proposed design implements discrete VLAN structures to achieve three critical engineering objectives:

- **Limitation of Broadcast Domains:** Multimedia and system discovery broadcasts are strictly confined within departmental boundaries. This preserves link bandwidth and optimizes the processing cycles of connected endpoint network interface cards (NICs).
- **Enforcement of Security Boundaries:** Departmental traffic is isolated at Layer 2. For instance, sensitive corporate data within the HR or Management networks is structurally invisible to the Sales broadcast domain, preventing unauthorized inter-departmental data sniffing.
- **Deterministic Policy Application:** By isolating departments into distinct logical subnets, network administrators can apply granular Quality of Service (QoS) priorities and Access Control Lists (ACLs) tailored to the specific operational needs of each business unit.

6.2 Structural Segmentation Strategy

To support specialized endpoint behaviours across the corporate infrastructure, the design establishes dedicated functional VLAN zones:

- **Departmental User VLANs (VLAN 10, 20, 40):** Standard corporate workspaces are mapped cleanly by organizational role (Sales, IT, and HR) to enforce strict operational compliance and tracking.
- **Dedicated Printer Networks (VLAN 30):** Network printing assets are grouped into a standalone broadcast domain. This design isolates heavy printer discovery traffic (such as Bonjour or WSD protocols) and prevents standard users from directly tampering with printer hardware management interfaces.
- **Management Infrastructure Zone (VLAN 50):** Administrative access interfaces for switches and routers are pinned to a dedicated management subnetwork. This separates network management traffic from standard user data, protecting the infrastructure from internal security threats.

- **Production Server Farm (VLAN 100):** Crucial enterprise resources (like HQ-SRV1) are placed inside a highly secure network zone. This allows engineers to strictly control access through targeted firewall and access list rules at the distribution layer.

6.3 Enterprise VLAN Allocation Table

The following matrix defines the formal allocation of virtual broadcast networks across both the Head Office campus and the remote Branch Office deployment.

Table 2: Organizational VLAN Allocation Matrix

Site Location	VLAN ID	Department/Zone Name	Operational Scope & Traffic Profile
Head Office	10	Sales	General outbound internet traffic and local intranet documentation access.
Head Office	20	IT Support	Elevated privilege access, network monitoring instrumentation, and system deployment.
Head Office	30	HQ Printers	Dedicated print queues, print server links, and hardware peripheral traffic.
Head Office	40	HR / Personnel	Secure employee database records, payroll access, and high-privacy operations.
Head Office	50	Management	Network device loopbacks, secure SSH terminal lines, and infrastructure control.
Head Office	100	Server Farm	Central corporate storage, local authentication engines, and shared business apps.
Branch Office	30	Branch Printers	Localized peripheral queues and printer management frameworks for remote operations.
Branch Office	40	Branch Users	Consolidated remote employee data traffic, local internet access, and WAN tunnel flows.

6.4 VISUAL ASSET INTEGRATION

To document the structural layout of these broadcast domains before implementation, the logical division of devices across the physical switches is mapped below.

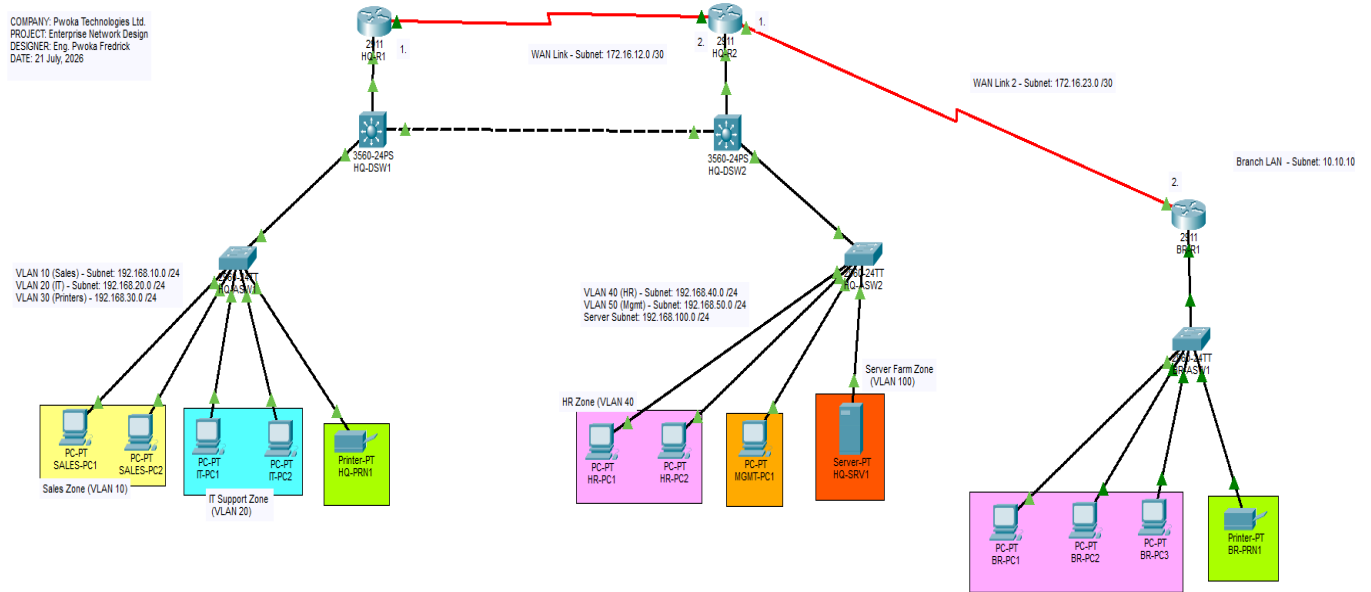


Figure 3: Logical VLAN Segmentation & Broadcast Domain Boundaries Map

7. IP Addressing Plan & WAN Design

7.1 IPv4 Addressing Strategy Justification

The logical addressing framework for Pwoka Technologies Ltd. is built using a structured classless IPv4 addressing model. To align with industry best practices, all internal campus and branch networks are allocated out of the **RFC 1918 Private Address Space** (the 192.168.0.0/16 and 10.0.0.0/8 blocks). This strategy ensures complete insulation from the public internet, preventing external entities from directly routing to internal corporate endpoints.

To optimize address conservation and match your planned topology exactly, this design implements two distinct masking models:

- **Local Area Networks (LANs):** Structured with a **Fixed-Length Subnet Mask (FLSM) of /24 (255.255.255.0)**. This configuration provides exactly 254 usable endpoint host identities per department. This capacity comfortably accommodates current operational staff while leaving a generous 90% margin for local hardware expansion.
- **Wide Area Network Links (WANs):** Engineered with a highly efficient **/30 mask (255.255.255.252)**. A /30 boundary yields exactly 2 usable host IP addresses. This matches point-to-point serial links perfectly and completely eliminates address wasting on the routing backbone.

7.2 Master Device IP Addressing Table

The following matrix serves as the definitive engineering reference for assigning network identities during the implementation phase. It includes all physical router ports, Layer 3 Switch Virtual Interfaces (SVIs), and representative enterprise end devices.

Table 3: Master IP Addressing Matrix

Device Hostname	Hardware Interface / SVI	Assigned IP Address	Subnet Mask	Default Gateway	Operational Role / Connected Segment
HQ-R1	Gi0/0	192.168.10.1	255.255.255.0	N/A	LAN Downlink to Campus Core Switch HQ-DSW1
HQ-R1	Se0/3/0 (DCE)	172.16.12.1	255.255.255.252	N/A	WAN Link 1 Interconnect to HQ-R2 Core
HQ-R2	Gi0/0	192.168.40.1	255.255.255.0	N/A	LAN Downlink to

					Campus Core Switch HQ-DSW2
HQ-R2	Se0/3/0 (DTE)	172.16.12.2	255.255.255.252	N/A	WAN Link 1 Interconnect back to HQ- R1
HQ-R2	Se0/3/1 (DCE)	172.16.23.1	255.255.255.252	N/A	WAN Link 2 Regional Interconnect to BR-R1
BR-R1	Gi0/0	10.10.10.1	255.255.255.0	N/A	LAN Downlink to Branch Access Switch BR- ASW1
BR-R1	Se0/3/0 (DTE)	172.16.23.2	255.255.255.252	N/A	WAN Link 2 Regional Interconnect back to HQ- R2
HQ-DSW1	VLAN 10	192.168.10.2	255.255.255.0	N/A	Active Layer 3 Gateway for Sales Network
HQ-DSW1	VLAN 20	192.168.20.2	255.255.255.0	N/A	Active Layer 3 Gateway for IT Support Network
HQ-DSW1	VLAN 30	192.168.30.2	255.255.255.0	N/A	Active Layer 3 Gateway for HQ Printers Network
HQ-DSW2	VLAN 40	192.168.40.2	255.255.255.0	N/A	Active Layer 3 Gateway for HR Network
HQ-DSW2	VLAN 50	192.168.50.2	255.255.255.0	N/A	Active Layer 3 Gateway for Managemen t Network
HQ-DSW2	VLAN 100	192.168.100.2	255.255.255.0	N/A	Active Layer 3 Gateway for Server Farm Zone

SALES-PC1	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.2	User Host Endpoint mapped to Sales Subnet
SALES-PC2	FastEthernet0	192.168.10.11	255.255.255.0	192.168.10.2	User Host Endpoint mapped to Sales Subnet
IT-PC1	FastEthernet0	192.168.20.10	255.255.255.0	192.168.20.2	User Host Endpoint mapped to IT Support Subnet
IT-PC2	FastEthernet0	192.168.20.11	255.255.255.0	192.168.20.2	User Host Endpoint mapped to IT Support Subnet
HQ-PRN1	FastEthernet0	192.168.30.10	255.255.255.0	192.168.30.2	Local Network Printing Peripheral Resource
HR-PC1	FastEthernet0	192.168.40.10	255.255.255.0	192.168.40.2	User Host Endpoint mapped to HR Subnet
HR-PC2	FastEthernet0	192.168.40.11	255.255.255.0	192.168.40.2	User Host Endpoint mapped to HR Subnet
MGMT-PC1	FastEthernet0	192.168.50.10	255.255.255.0	192.168.50.2	Corporate Administrative Command Station
HQ-SRV1	FastEthernet0	192.168.100.10	255.255.255.0	192.168.100.2	Centralized Active Directory / File Host Node
BR-PC1	FastEthernet0	10.10.40.10	255.255.255.0	10.10.40.254	Remote User Station mapped to Branch Network

BR-PC2	FastEthernet0	10.10.40.11	255.255.255.0	10.10.40.254	Remote User Station mapped to Branch Network
BR-PC3	FastEthernet0	10.10.40.12	255.255.255.0	10.10.40.254	Remote User Station mapped to Branch Network
BR-PRN1	FastEthernet0	10.10.30.10	255.255.255.0	10.10.30.254	Remote Branch Office Peripheral Printer

7.3 Wide Area Network (WAN) Link Profiles

The point-to-point Wide Area Network links simulate lease-line telco serial paths using dedicated Data Communications Equipment (DCE) clocking mechanisms:

- **WAN Link 1 (Head Office Backbone Core):** Deployed across subnet 172.16.12.0/30. HQ-R1 assumes the master clocking profile (**DCE rate 128000**) on port Se0/3/0, connecting directly to the downstream Data Terminal Equipment (DTE) port on HQ-R2.
- **WAN Link 2 (Head Office to Remote Branch):** Deployed across subnet 172.16.23.0/30. HQ-R2 assumes the master clocking profile (**DCE rate 128000**) on port Se0/3/1, extending across the regional boundary to terminate at the branch edge interface of BR-R1.

7.4 Visual Asset Integration

To isolate the logical parameters of the WAN backbone, a simplified physical and logical network routing overview is mapped below.

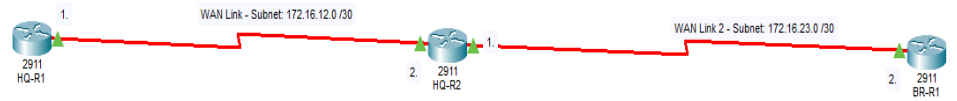
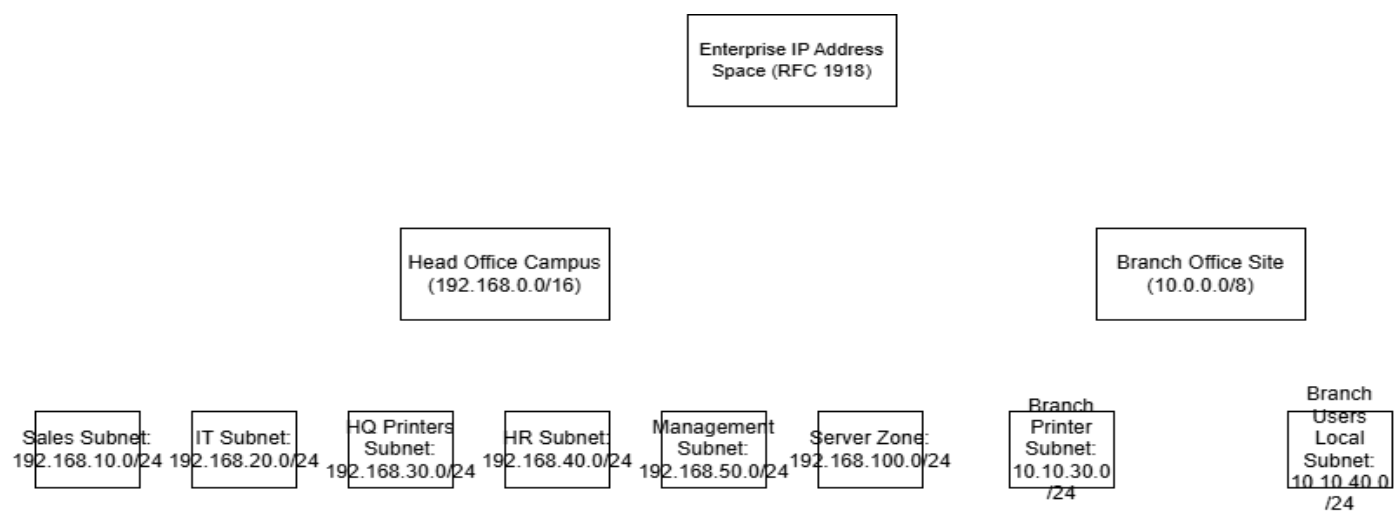


Figure 5. Wide Area Network (WAN) Backbone Link & Routing Topology Diagram

Figure 6: logical IP Addressing Hierarchy Blueprint



8. REDUNDANCY & HIGH-AVAILABILITY DESIGN

8.1 ARCHITECTURAL RESILIENCY STRATEGY

The proposed network design for Pwoka Technologies Ltd. is specifically engineered to eliminate single points of failure within the campus core. In modern enterprise environments, data link outages or hardware failures at the distribution layer can lead to costly operational downtime. This design implements structured physical redundancy and automated failover protocols at both Layer 2 and Layer 3 to ensure uninterrupted corporate communications.

8.2 Layer 2 Bandwidth Aggregation (LACP EtherChannel)

To prevent network bottlenecks between the core distribution switches, the design includes a high-capacity logical backbone link connecting **HQ-DSW1** and **HQ-DSW2**.

- **Protocol Choice:** Industry-standard **Link Aggregation Control Protocol (LACP - IEEE 802.3ad)** is chosen over proprietary solutions to ensure open, vendor-neutral compatibility.

- **Interface Bundling:** Physical ports **Gigabit Ethernet 0/2** on both switches are aggregated into a single logical channel designated as **Port-Channel 1**.
- **Operation Mode:** Configured in **Active Negotiation Mode**, meaning both switches actively initiate control packets to build and maintain the bundle.
- **Engineering Benefits:** This trunk configuration doubles the physical throughput capacity of the switch backbone while offering seamless link failover. If one connection drops, traffic instantly shifts to the remaining active link within milliseconds without dropping user sessions.

8.3 Layer 3 Gateway Redundancy (HSRP Design)

To maintain constant uptime for end-user workstations, the design includes first-hop gateway redundancy at the Head Office using the **Hot Standby Router Protocol (HSRP)**.

- **The High-Availability Problem:** If a department switch or router interface acting as a default gateway goes offline, all user PCs on that subnet lose access to external networks.
- **The HSRP Solution:** By configuring HSRP across the Layer 3 switches, **HQ-DSW1** and **HQ-DSW2** share a single, shared **Virtual IP Address** for each subnetwork. End-user workstations point to this virtual address as their default gateway instead of the physical interface of an individual switch.
- **Operational States:**
 - Active Router:** One switch is assigned a higher HSRP priority value (e.g., 105) to actively route all outbound corporate packets under normal operating conditions.
 - Standby Router:** The secondary switch maintains a default priority value (e.g., 100), listening for automated keepalive messages. If the active switch fails, the standby switch instantly takes over the virtual gateway identity. This transparent transition keeps user applications connected without requiring manual intervention from IT support.

8.4 Loop Mitigation (Spanning Tree Protocol Baseline)

Because redundant physical cable loops exist between the access layer and distribution switches, the infrastructure relies on **Spanning Tree Protocol (STP - IEEE 802.1D/802.1w)** to prevent disastrous broadcast loops.

- **Path Optimization:** The design assigns primary and secondary **STP Root Bridge** priorities to the distribution switches. **HQ-DSW1** is designated as the primary root bridge for the left-side campus subnets (VLANs 10, 20, 30), while **HQ-DSW2** operates as the primary root bridge for the right-side subnets (VLANs 40, 50, 100).

- **Edge Port Fast-Forwarding:** To improve user connectivity speeds at the edge, all end-device access ports are configured with **Spanning-Tree Port Fast**. This allows user PCs and printers to bypass standard listening and learning delays, safely transitioning ports directly to the forwarding state within one second of physical link attachment.

9. ROUTING STRATEGY

9.1. DYNAMIC ROUTING PROTOCOL SELECTION (EIGRP AS 100)

To establish seamless, multi-site connectivity for Pwoka Technologies Ltd., the core network infrastructure utilizes a dynamic routing protocol rather than restrictive static routing. The **Enhanced Interior Gateway Routing Protocol (EIGRP)** is selected as the primary routing engine for this enterprise environment.

- **Autonomous System Number:** The design implements **EIGRP Autonomous System 100 (AS 100)** consistently across all corporate routing nodes.
- **Engineering Justification:** EIGRP is chosen over alternative distance-vector or link-state protocols due to its rapid convergence metrics, low processing overhead, and use of the **Diffusing Update Algorithm (DUAL)**. DUAL guarantees a loop-free topology while computing fallback routes (Feasible Successors) in advance. This allows the network to reroute traffic around a failed WAN link within milliseconds without needing a full protocol recalculation.

- **Classless Flexibility:** EIGRP natively supports Classless Inter-Domain Routing (CIDR) and Variable Length Subnet Masking (VLSM), which ensures precise matching and processing of our /24 departmental blocks and /30 point-to-point wide-area connections.

9.2 Inter-VLAN Routing Architecture

To facilitate communication between separate internal departments at the Head Office, routing is handled directly at the distribution layer using the wire speed hardware engines of the **Cisco 3560 Multilayer Switches**.

- **Switch Virtual Interfaces (SVIs):** Rather than implementing an inefficient "Router-on-a-Stick" sub interface architecture, this design configures unique **Layer 3 SVIs** directly on **HQ-DSW1** and **HQ-DSW2**.
- **The Layer 3 Boundary:** By executing the IP routing command globally on the multilayer switches, they switch local traffic at Layer 2 within the same department while routing cross-departmental packets at Layer 3 at full hardware wire speed. This design offloads the internal traffic processing burden entirely from the core edge routers (HQ-R1 and HQ-R2), reserving their WAN interfaces exclusively for external and inter-site traffic transport.

9.3 Core WAN Routing Propagation

To advertise network routes across the entire enterprise, the layout coordinates route distribution between the campus switching core and the remote branch office over the serial backbones:

1. **Head Office Core Redistribution:** The Layer 3 multilayer switches (HQ-DSW1 and HQ-DSW2) establish EIGRP neighbour relationships with their adjacent core routers (HQ-R1 and HQ-R2) over their dedicated physical Gigabit Ethernet links.
2. **WAN Path Advertisement:** The routers peer over the /30 serial interfaces. HQ-R1 and HQ-R2 advertise the internal campus network subnets (192.168.0.0/16) across the wide area network to the branch edge.
3. **Branch Site Learning:** The branch router (BR-R1) receives these advertisements over its Se0/3/0 interface and populates its local routing table. Concurrently, BR-R1 advertises the remote branch local user networks (10.10.40.0/24) and printer networks (10.10.30.0/24) back to the Head Office core.

This configuration guarantees that any employee sitting at a desk in the Branch Office can successfully ping, authenticate, and communicate with the centralized HQ-SRV1 server farm at the Head Office.

9.4 Routing Optimization Metrics

To ensure the routing topology operates at peak efficiency, the configuration implements the following optimization baselines:

- **Manual Route Summarization:** To protect the narrow bandwidth links of our wide-area connections, core routers summarize internal routes into a single clean summary boundary (192.168.0.0/16) before sending them across the WAN. This practice keeps the remote branch router's routing table small and prevents local network changes at the Head Office from causing routing updates over the WAN.
- **Disable Automatic Summarization:** The explicit execution of the no auto-summary command is required within the EIGRP routing instance configuration. This ensures the protocol respects classless subnets and does not consolidate them back into standard Class A, B, or C boundaries, preventing routing black holes.

10. SECURITY & INFRASTRUCTURE HARDENING STRATEGY

10.1 ADMINISTRATIVE CONTROL PLANE PROTECTION

To secure the physical and logical assets of Pwoka Technologies Ltd., this design establishes a comprehensive hardening baseline across all core devices. Relying on default device configurations leaves a network highly vulnerable to malicious internal access and eavesdropping. This design implements three key layers of control plane security:

- **Encrypted Remote Management (SSHv2):** Insecure clear-text Telnet access is completely blocked across all virtual terminal lines (line vty 0 4). It is replaced with **Secure Shell Version 2 (SSHv2)** using **1024-bit RSA crypto keys**, which encrypts all remote administrative data sessions over the network.

- **Granular Password Encryption:** Global password encryption (service password-encryption) is turned on across all routers and switches. This alters all clear-text strings stored in the configuration files, protecting administrative credentials from shoulder-surfing attempts during live console sessions.
- **Role-Based Local Authentication:** Management line access requires a localized administrator credential challenge (login local), binding access privileges strictly to verifiable system accounts.

10.2 Layer 2 Edge Access Hardening

Because the physical access layer switches are closest to untrusted user devices, the design applies targeted security measures at the network edge to eliminate common Layer 2 security threats:

- **Port Security Infrastructure:** To stop MAC-address flooding attacks, all active edge access interfaces are locked down using **Cisco Port Security**. Ports are configured with a maximum allowance of **one (1) unique MAC address**, utilizing **Sticky Learning** behaviour to dynamically bind and save the user's MAC address directly to the running configuration file.
- **Violation Enforcement:** If an unauthorized device or rogue network hub is plugged into a secured switch port, the port security engine forces the interface into an immediate **Shutdown state**. This action cuts off link access, logs a syslog message for IT support, and requires manual administrative intervention to bring the port back online.
- **Mitigation of Layer 2 Spanning-Tree Exploits:** To protect the core root bridge topology from rogue device manipulation, **BPDUGuard** is turned on globally across all end-user edge access interfaces. If a rogue switch is plugged into a user port and sends a Bridge Protocol Data Unit (BPDU), the interface instantly shuts down to protect the spanning-tree topology.

10.3 Layer 2 Backbone Infrastructure Hardening (DHCP Snooping)

To protect against rogue network service attacks, the design includes layer 2 infrastructure validation mechanisms across all access switches:

- **DHCP Snooping Baseline:** To stop malicious users from running unauthorized DHCP servers that could intercept corporate traffic (Man-in-the-Middle attacks), the design turns on **DHCP Snooping** across all user VLANs.
- **Interface Classification boundaries:**
 - Trusted Interfaces:** Core switch trunks and interfaces directly connected to authorized distribution layer gateways or legitimate DHCP relay paths are explicitly flagged as **Trusted**.

Untrusted Interfaces: All general end-user client access ports are classified as **Untrusted** by default. If a rogue device connected to an untrusted port attempts to reply with malicious DHCP offer messages, the switch drops the packet instantly and shuts down the interface.

10.4 Traffic Filtering Strategy (Access Control Lists)

To control communication flows across different departments, the design uses targeted **Access Control Lists (ACLs)** applied at the Layer 3 boundaries of the distribution switches.

- **The Security Rule Principle:** By default, internal Inter-VLAN routing allows all subnets to communicate freely. This design modifies that behaviour by applying the security principle of **Least Privilege**.
- **Targeted Department Isolation:**
 - Standard User Restrictions:** Standard department workstations (such as Sales or HR hosts) are blocked from initiating direct remote connections to the core infrastructure management networks.
 - Server Farm Protection:** Traffic traveling into the production server subnetwork (192.168.100.0/24) is filtered using **Extended Named ACLs**. This configuration allows users to access specific business application ports (like HTTP/HTTPS or File Services) while blocking all unauthorized network probing tools and administrative management traffic.

11. Network Growth & Scalability Plan

11.1 Proactive Expansion Architecture

A major flaw in many student projects is looking only at the immediate requirements while ignoring future corporate growth. The proposed infrastructure for Pwoka Technologies Ltd. avoids this pitfall by building room for expansion directly into the core design. By implementing structured logical addressing and modular hardware layers, the organization can scale its operations smoothly over the next 5 to 10 years without requiring a costly network redesign.

11.2 Logical Subnet Capacity Analysis

The choice of standard classless /24 subnet masking across all internal local networks provides massive room for team growth:

- **Host Margin Capability:** Each department network block supports exactly **254 usable endpoint IP addresses**.
- **Current vs. Future Utilization:** Current team deployments use fewer than 10 physical endpoints per subnetwork block. This leaves a comfortable **90% expansion buffer** within every single configured VLAN.
- **Seamless Scalability:** The company can add new user workstations, corporate network printers, or internal department hardware assets instantly. New devices simply need to be assigned an available static IP within that department's subnet range and plugged into an open access switch port.

11.3 Modular Architectural Scalability

Because the design relies on the modular Cisco Hierarchical Model, adding new business locations or technology services is straightforward:

- **Adding New Departments:** If the company opens a new department (such as Finance or Legal), network engineers can assign a new, sequential VLAN ID (e.g., VLAN 60) and allocate an unused subnetwork block (such as 192.168.60.0/24) without changing any existing department configurations.
- **Adding Extra Switches:** If an access layer switch fills up, engineers can insert an additional access switch into the rack and connect it to the distribution layer switches (HQ-DSW1 and HQ-DSW2) using standard redundant trunk links. This increases port capacity without disrupting existing network traffic.
- **Future Service Integration:** The underlying network foundation is engineered to support future enterprise upgrades, including:
 - **Voice over IP (VoIP):** Dedicated Voice VLANs can be introduced to provide clean quality-of-service (QoS) routing for corporate IP phones.
 - **Wireless Local Area Networks (WLANs):** Cisco Wireless LAN Controllers (WLCs) and lightweight access points can be deployed into a dedicated management VLAN to give employees secure wireless access.
 - **Cloud Connectivity:** The core routers can be upgraded with dedicated VPN tunnels or leased-line extensions to establish secure, direct links to cloud providers like AWS or Microsoft Azure.

12. EXPECTED OPERATIONAL OUTCOMES & CONCLUSION

12.1. POST-IMPLEMENTATION BEHAVIORAL EXPECTATION

Once this design proposal successfully transitions into the implementation phase within Cisco Packet Tracer, the corporate network must operate according to strict, deterministic engineering parameters. Network verification testing should confirm the following baseline behaviours:

- **Localized Intra-VLAN Forwarding:** Endpoint host devices connected within the same functional department (such as SALES-PC1 and SALES-PC2) must achieve immediate, wire-speed communication at Layer 2 without involving the Layer 3 routing engine.
- **Deterministic Inter-VLAN Transit:** Cross-departmental traffic packets must travel securely through the active Layer 3 Switch Virtual Interfaces (SVIs) on the multilayer switches (HQ-DSW1 and HQ-DSW2), where all active security policies and Access Control Lists (ACLs) are enforced.
- **Resilient Gateway Path Failover:** If an active distribution switch suffers a sudden hardware failure, the **Hot Standby Router Protocol (HSRP)** state engine must transparently transition the standby switch into the active gateway role. This failover should occur within three seconds, ensuring corporate endpoints do not lose connection to the network backbone.
- **Dynamic Wide-Area Network Convergence: EIGRP AS 100** must successfully establish neighbour relationships across the /30 point-to-point serial links. This protocol will dynamically populate the routing tables of HQ-R1, HQ-R2, and BR-R1, allowing remote Branch Office workstations to securely access central resources like the HQ-SRV1 server farm.
- **Edge Port Hardening Enforcement:** The **Cisco Port Security** and **DHCP Snooping** engines must actively monitor edge connections. If an unauthorized device or rogue DHCP server is connected to an un-trusted interface, the switch must instantly force that specific port into an error-disabled shutdown state to protect the rest of the enterprise network.

12.2 Project Conclusion

The proposed network design proposal for **Pwoka Technologies Ltd.** delivers a comprehensive, highly secure, and fault-tolerant network framework that fully supports the organization's multi-site business operations. By moving away from a vulnerable flat network topology and adopting the **Cisco Hierarchical Design Model**, this architecture successfully eliminates single points of failure, minimizes broadcast network congestion, and enforces strict security baselines at the network edge.

This document serves as a complete, verified logical blueprint. It provides a structured foundation that guarantees a smooth implementation phase within Cisco Packet Tracer, satisfying all academic criteria for the CCNA3 enterprise curriculum.

